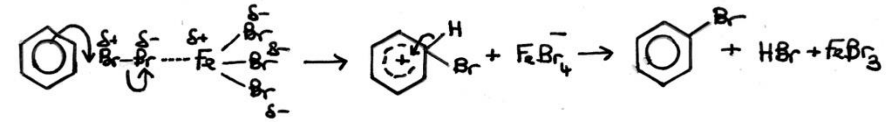
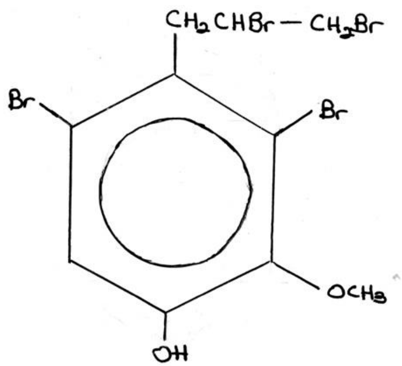
Question				Marking details	Marks available					
					AO1	AO2	AO3	Maths	Prac	Total
10.	(a)			•  accept valid variations on this mechanism • The Br–Br bond becomes polarised δ+ and δ− by • Electron movement / attraction from the π electron system • Polarisation from the electron deficient iron atom in $FeBr_3$ • Mechanism shows the catalyst $FeBr_3$ regeneration • Mechanism shows the production of the co-product, hydrogen bromide • In the absence of the catalyst there is reduced / little induced polarisation of the bromine molecule by the π electron system **5-6 marks** Good account including detailed comments about the mechanism in both the presence and absence of the catalyst. *The candidate constructs an articulate, integrated account, which shows sequential reasoning. The answer fully addresses the question with no irrelevant inclusions or significant omissions.* *The candidate uses scientific conventions and vocabulary appropriately and accurately*. **3-4 marks** Most of the information has been provided but the account has less detailed comments about the mechanism in both the presence and absence of the catalyst. *The candidate constructs an account correctly linking some relevant points showing some reasoning. The answer addresses the question with some omissions. The candidate usually uses scientific conventions and vocabulary appropriately and accurately*.		3	3			6

Question				Marking details	AO1	AO2	AO3	Maths	Prac	Total
				1-2 marks Some of the information has been provided but the account lacks details about the mechanism. *The candidate makes some relevant points showing limited reasoning. The answer addresses the question with significant omissions. The candidate makes limited use of scientific conventions and vocabulary.* **0 marks** *The candidate does not make any attempt or give a relevant answer worthy of credit.*						
	(b)	(i)		blue / purple colouration	1				1	1
		(ii)		award (1) for either of following goes from yellow-brown → colourless goes from yellow-brown → white (precipitate)	1				1	1

Question				Marking details	Marks available					
					AO1	**AO2**	**AO3**	**Maths**	**Prac**	**Total**
		(iii)		% bromine by mass = 66.4 (1) dividing by A_r (1) then dividing each by the smallest figure gives ratio leading to empirical formula of $C_5H_5Br_2O$ (1) compound contains two oxygen atoms / 10 carbon atoms so molecular formula is $C_{10}H_{10}Br_4O_2$ (1) possible structure is  (1)	1	1 1	1 1	3		5
		(iv)		eugenol is **acidic** (phenol) and will **react with NaOH to give a salt** (which is soluble in the aqueous layer) (1) eugenyl ethanoate is an **ester** and is **not soluble in the aqueous layer** (1)			2			2
				Total question 10	**3**	**5**	**7**	**3**	**2**	**15**